

ABDELRAHMAN ALORDY

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RESEARCH INTERESTS

- Robotic Manipulation in Unstructured Environments
- Deep Learning for Perception & Control
- Sim-to-Real Transfer & Domain Adaptation
- Reinforcement Learning & Sequential Decision-Making
- Multi-Modal Learning & Embodied AI
- Machine Learning Systems & Intelligent Software Engineering

EDUCATION

B.Sc. in Computers and Information (Software Engineering)

Cairo University, Faculty of Computers and Artificial Intelligence

Oct. 2020 – May 2025

Giza, Egypt

- GPA: 3.4/4.0
- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Computer Architecture, Operating Systems, Algorithms & Data Structures, Probability & Statistics

Visiting Student (Study Abroad Semester)

University of Louisville, J. B. Speed School of Engineering

Aug. 2023 – Dec. 2023

Louisville, KY, USA

- **Coursework:** Embedded Systems, Data Analytics, Data Analysis and Decision Making

RESEARCH EXPERIENCE

AI & Robotics Manipulation Research Intern

Singapore Institute of Manufacturing Technology (SIMTech), A*STAR

Supervisor: Dr. Zhu Haiyue – [GitHub Repository](#) – [Video](#)

Sep. 2024 – Dec. 2024

Singapore

- Developed a deep-learning-based robotic grasping system on a UR10e arm using ROS2, MoveIt2, RViz, and Intel RealSense for real-time object detection, pose estimation, and grasp execution in cluttered environments.
- Resolved sim-to-real, sensor noise, and depth inconsistencies challenges via multi-sensor calibration and grasp filtering.
- Designed a microservices-based control pipeline using Python for modular perception, planning, and execution, boosting reliability and reducing latency under hardware constraints.
- Conducted grasp success experiments under varying conditions, generating datasets for improving the AnyGrasp model and identifying real-world transfer limitations.

ML & Scientific Computing Research Intern

Meshcheryakov Laboratory of Information Technologies (MLIT), JINR

Supervisor: Dr. Slavomir Hnatič

Aug. 2024 – Sep. 2024

Dubna, Russia

- Implemented ML models for classification/regression on large datasets using Python, NumPy, SciPy, and PyTorch for scientific analysis for the Multi-Purpose Detector project.
- Collaborated with physicists to enhance primary vertex detection for the MPD, achieving a 9% accuracy gain over baselines.
- Explored optimization, uncertainty handling, and noise reduction in physical-measurement data.
- Fine-tuned and evaluated C++ event reconstruction algorithms, improving accuracy and data flow efficiency.

PROFESSIONAL EXPERIENCE

Software Engineer

Iskraemeco

Jun. 2025 – Aug. 2025

Cairo, Egypt

- Optimized .NET/Python backend for integrated industrial metaverse, focusing on scalable design and data pipeline.
- Managed sensor/operational data with automation scripts, emphasizing reliability and latency.

Software Engineer Intern

Halliburton

Jul. 2023 – Aug. 2023

Cairo, Egypt

- Implemented Java/Spring Boot services for data handling, task tracking, and reports with PostgreSQL/Docker.
- Optimized real-time processing and scheduling for performance and modularity.

SELECTED PROJECTS

Emotion Recognition System (Python, NumPy, Pandas, TensorFlow, Keras)

- Built a CNN-based model on FER2013, improving accuracy from 64% to 70% via custom preprocessing and augmentation.
- Implemented early stopping, batch normalization, and hyperparameter tuning, reducing validation loss by 18%.

Object Recognition and Tracking Systems (Python, OpenCV, TensorFlow)

- Developed CIFAR-10 model at 85% accuracy after tuning with dropout and batch normalization (loss -12%).
- Designed real-time vehicle tracking at 92% accuracy/18 FPS using contour detection and background subtraction.

CampusEats (React Native, Node.js, MySQL)

- Built a full-stack university dining app with personalized recommendations, real-time data management, and an MVC-based, scalable architecture following SOLID principles and design patterns.

TECHNICAL SKILLS

Programming: Python, C++, Java, C#, SQL
ML & Robotics: TensorFlow, PyTorch, NumPy, SciPy, Scikit-learn, OpenCV, ROS2, MoveIt2, RealSense SDK, RViz, Gazebo
AI & Learning: Deep Learning, Reinforcement Learning Basics, Sim-to-Real Transfer, Dataset Curation & Augmentation
Software Engineering: REST APIs, Microservices, MySQL, CI/CD, Docker, Git, Jira, Ubuntu 22.04

SCHOLARSHIPS & AWARDS

USAID Scholars Activity – Full Bachelor’s Scholarship <i>United States Agency for International Development (USAID)</i>	<i>Oct. 2020 – Jan. 2025</i> <i>Egypt</i>
Singapore International Pre-Graduate Award (SIPGA) <i>Agency for Science, Technology and Research (A*STAR)</i>	<i>Sep. 2024 – Dec. 2024</i> <i>Singapore</i>
Student Advanced Research Training (START) <i>Joint Institute for Nuclear Research (JINR)</i>	<i>Aug. 2024 – Sep. 2024</i> <i>Russia</i>
U.S. Study Abroad Program <i>United States Agency for International Development (USAID)</i>	<i>Aug. 2023 – Dec. 2023</i> <i>USA</i>

TEST SCORES

IELTS Academic (Nov. 18, 2025)

- Overall: 8.0 (Reading: 8.5 | Listening: 7.5 | Writing: 8.0 | Speaking: 8.0)